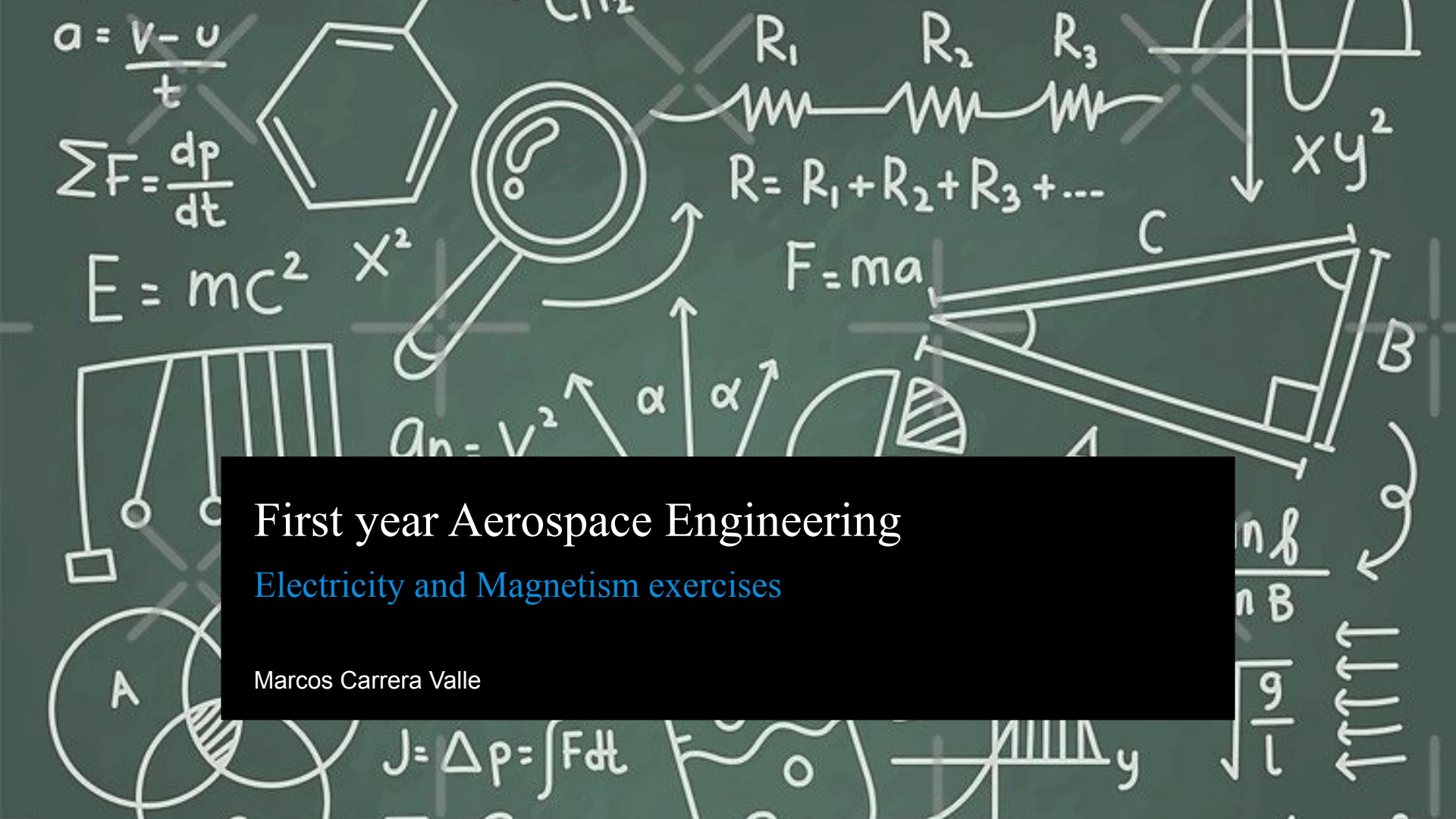


The background is a dark green chalkboard filled with various white hand-drawn diagrams and mathematical formulas. These include:

- Formulas: $a = \frac{v-u}{t}$, $\Sigma F = \frac{dp}{dt}$, $E = mc^2$, x^2 , $F = ma$, $R = R_1 + R_2 + R_3 + \dots$, $a_n = v^2$, $J = \Delta p = \int F dt$, $\sqrt{\frac{g}{l}}$, and xy^2 .
- Diagrams: A benzene ring, a magnifying glass over a question mark, three resistors in series, a sine wave, a pendulum, a triangle with sides labeled A, B, and C, a circular sector, a spring, and a series of horizontal arrows.

First year Aerospace Engineering

Electricity and Magnetism exercises

Marcos Carrera Valle

- 1) A resistor dissipates 2,400 Joules of thermal energy in exactly 2 minutes. During this time, a total charge of 600 Coulombs flows through the resistor. Calculate the resistance of the component in Ohms.
- 2) A smartphone battery is rated at 4,000 mAh (milliamp-hours) and operates at a constant voltage of 3.7 Volts. If the phone draws a constant current of 200 mA, calculate the total electrical energy stored in the battery (in Joules) and how long (in hours) the battery will last.
- 3) During a massive storm, a lightning bolt transfers 15 Coulombs of charge from a cloud to the ground in a duration of 0.0005 seconds. The potential difference (voltage) between the cloud and the ground is 100,000,000 Volts. Calculate the average current of the strike and the total power delivered during that split second.
- 4) An electric water heater is connected to a 240 V mains supply. It has an internal resistance of 12 Ohms. Calculate how many Coulombs of charge flow through the heating element if it is left running for 1 hour.
- 5) A power station sends current through a long cable with a total resistance of 5 Ohms. The current flowing through the cable is 100 Amps. Calculate the total energy lost as heat (in Megajoules) over a period of 24 hours.

6) A device claims to consume 50 Watts of power while running on a 10 Volt battery. However, the fuse protecting the device will blow if the charge flow exceeds 120 Coulombs per minute. Will the fuse blow? Prove your answer by calculating the current.

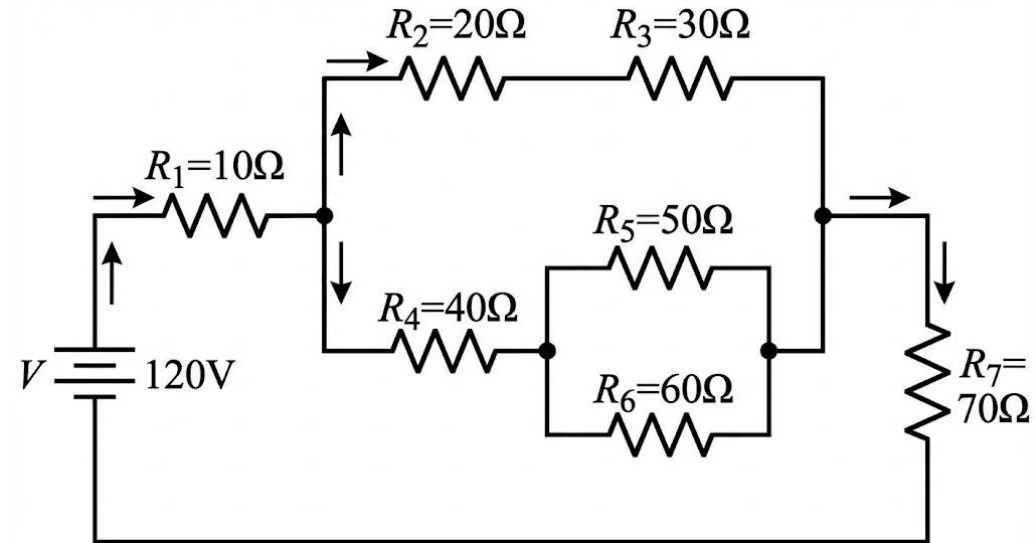
7) A variable resistor is adjusted such that the current flowing through it is 2 Amps. In 10 seconds, the resistor dissipates 400 Joules of energy. Calculate the voltage drop across the resistor and its resistance value.

8) A tiny circuit runs for 1 minute with a constant current of 0.0000032 Amps (3.2 microamps). Given that the charge of a single electron is 1.6×10^{-19} C, calculate the total number of electrons that passed through the circuit and the total energy consumed if the voltage was 5 V.

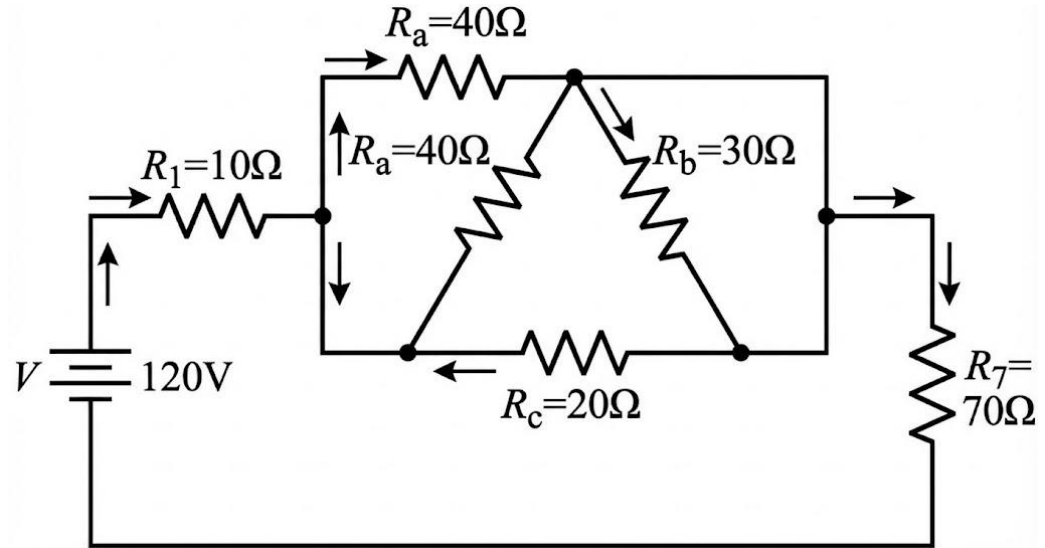
9) An electric motor connected to a 120 V source draws 5 Amps. The motor runs for 8 hours. If electricity costs \$0.15 per kilowatt-hour (kWh), calculate the total cost to run the motor.

10) System A operates at 200 V with a resistance of 100 Ohms. System B operates at 50 V with a resistance of 5 Ohms. Which system processes more charge in 1 hour, and which system consumes more power?

- 11) Find the equivalent resistance of this circuit
- 12) Find the power dissipated through resistor R_6

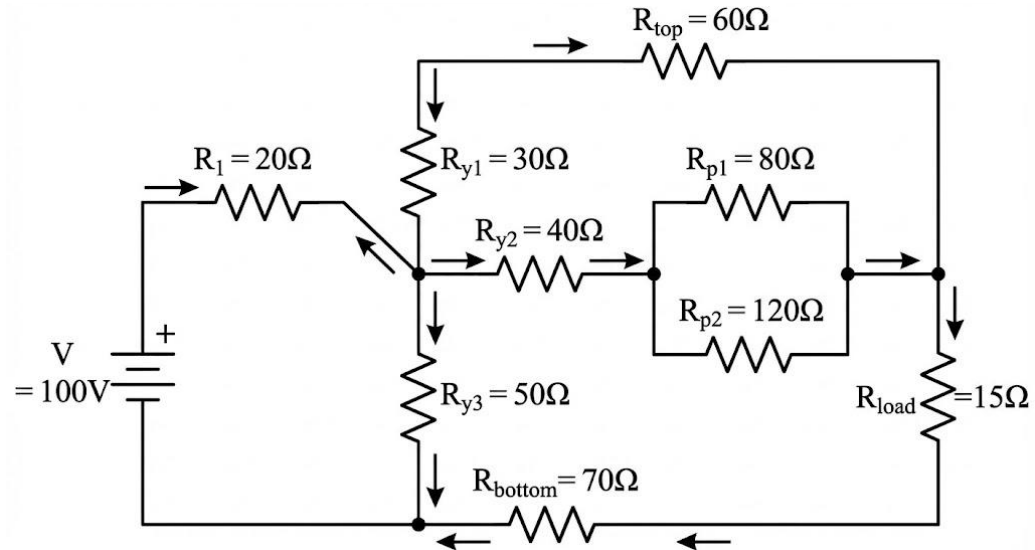


- 13) Find the equivalent resistance of this circuit
14) Find the power dissipated through resistor R_c

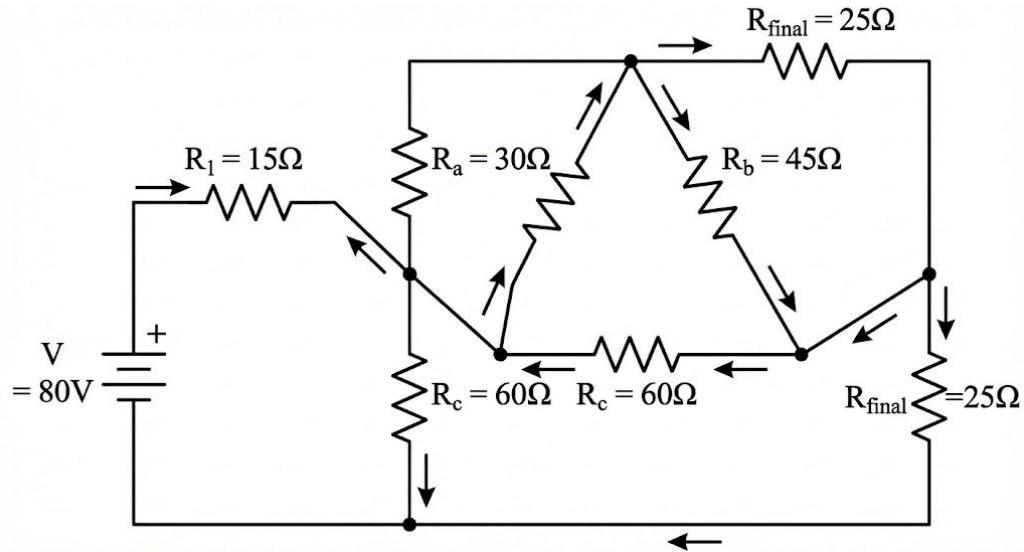


15) Find the equivalent resistance of this circuit

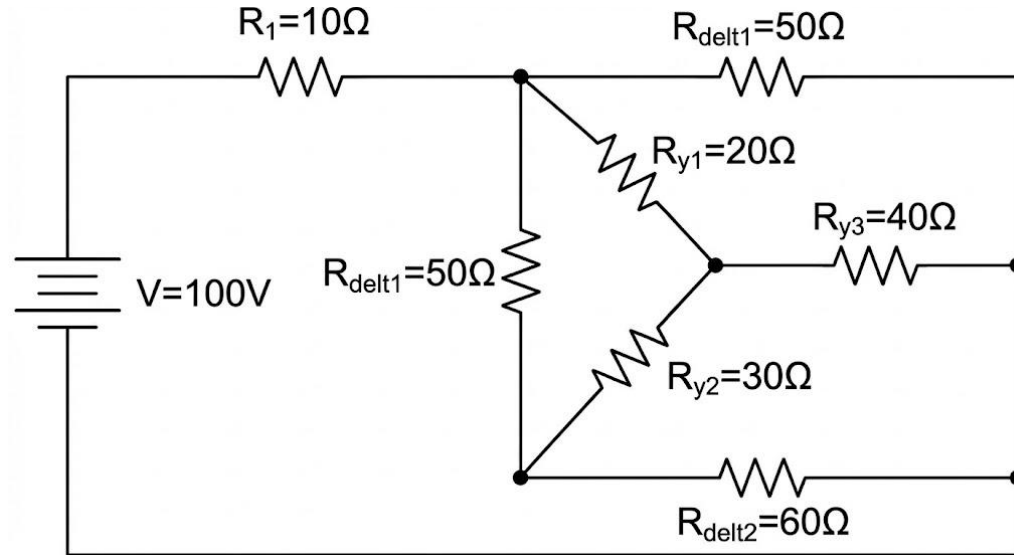
16) Find the current through resistor R_{y3}



- 17) Find the equivalent resistance of this circuit
18) Find the power dissipated through resistor



- 19) Find the equivalent resistance of this circuit
20) Find the power dissipated through resistor R_{y2}



21) Two identical small pith balls, each with mass m and charge $+q$, hang from a single point by two non-conducting strings, each having length L . Because they repel each other electrostatically, the strings hang at an angle θ with the vertical when the system is balanced. Using this information, derive the formula for the magnitude of the charge q using m , L , θ , and the Coulomb constant k .

22) A small bead with charge $+q$ and mass m is connected to one end of a non-conducting horizontal spring. This spring has a spring constant k and a natural, relaxed length of L_0 . The opposite end of the spring is attached to a wall. Another identical charge $+q$ is fixed to the wall at the exact spot where the spring connects. The entire setup sits on a frictionless table. If the spring stretches by a distance x to reach a new steady length L , write down the equation balancing the forces that would let you solve for x .

23) Two small spheres, each with a mass of 0.005 kg , are placed inside a vertical glass tube. The bottom sphere is locked in place and carries a charge of $+2.0 \text{ microCoulombs}$. The top sphere is free to move and carries a charge of $+3.0 \text{ microCoulombs}$. At what height h above the bottom sphere will the top sphere float in a stable equilibrium? (Assume $g = 9.8 \text{ m/s}^2$ and $k = 9.0 \times 10^9 \text{ N m}^2/\text{C}^2$).

24) Two positive point charges, $+Q$ and $+9Q$, are locked in place at a distance d apart. A third charge q needs to be placed somewhere along the line connecting the two charges so that the entire system is in equilibrium (meaning the net force on every single charge is zero). Figure out the required position, magnitude, and sign of this third charge q .

25) In a simplified model of the hydrogen atom, an electron (mass m_e , charge $-e$) orbits a proton (mass m_p , charge $+e$) in a circle with radius r . Assuming the proton stays still and the only force acting between them is the electrostatic force, derive the formula for the kinetic energy of the electron using k , e , and r .

26) A long, straight wire is aligned perfectly North-South and carries a strong steady DC current flowing from South to North. A magnetic compass is placed directly below the wire. Ignoring the effect of the Earth's magnetic field, in which compass direction (North, South, East, or West) will the North pole of the needle point?

27) A vertical wire carries a current flowing vertically upwards. An electron beam is fired parallel to the wire, also moving upwards, at a distance of 10 cm from the wire. Does the electron beam bend towards the wire, bend away from the wire, or continue in a straight line?

28) Two long parallel wires are separated by a distance of 1.0 meter. Wire A carries a current of 2.0 Amps, and Wire B carries a current of 6.0 Amps. Both currents flow in the same direction. At what specific distance from Wire A is the net magnetic field equal to zero?

29) A Geiger counter detects a certain magnetic field strength B at a distance d from a long straight wire carrying current I . If the current in the wire is tripled ($3I$) and the detector is moved to a distance three times further away ($3d$), what is the new magnetic field strength in terms of B ?

30) Two insulated long wires cross each other at 90 degrees without electrical contact. Wire 1 is vertical and carries current Up. Wire 2 is horizontal and carries current to the Right. Consider the four quadrants created by this cross. In which two quadrants do the magnetic fields from the two wires oppose each other (subtract), and potentially cancel out to zero?

31) A proton (charge $q = +e$) enters a magnetic field with a velocity vector $\mathbf{v} = (200 \mathbf{i} + 300 \mathbf{k})$ m/s. The magnetic field is uniform and points entirely along the x-axis: $\mathbf{B} = 0.5 \mathbf{i}$ Tesla. Calculate the magnitude and direction (using vector notation) of the Lorentz force acting on the proton at this instant.

32) A rigid wire carries a current of 2.0 Amps. The wire is bent into an "L" shape. Segment 1 runs along the x-axis from $x=0$ to $x=3$ meters. Segment 2 runs along the y-axis from $y=0$ to $y=4$ meters. The entire wire sits in a uniform magnetic field $\mathbf{B} = 2.0 \mathbf{k}$ Tesla (pointing out of the page). Calculate the magnitude of the total net force acting on the entire bent wire.

33) An electron (charge $q = -1.6 \times 10^{-19}$ C) is moving vertically downward (in the negative $\mathbf{j}$ direction) at a speed of 3×10^6 m/s. It encounters a magnetic field pointing North (assume North is positive $\mathbf{k}$ and East is positive $\mathbf{i}$). Determine the vector direction of the magnetic force. Does it point East, West, North, or South?

34) A metal rod of mass 0.2 kg and length 1.0 meter slides without friction on two parallel vertical rails. A uniform magnetic field of 0.5 Tesla points horizontally perpendicular to the rod (into the page, negative k direction). A battery connects to the rails, sending a current I through the rod. Calculate the magnitude of the current I required to keep the rod suspended motionless against gravity ($g = 9.8 \text{ m/s}^2$), and state the direction the current must flow through the rod (Left-to-Right or Right-to-Left).

35) A straight wire of length 2 meters lies in the xy-plane, oriented along the line $y = x$ (pointing in the direction of $\mathbf{i} + \mathbf{j}$). It carries a current of 5 Amps. The magnetic field is not aligned with any single axis; it points diagonally: $\mathbf{B} = (0.3 \mathbf{i} - 0.3 \mathbf{j})$ Tesla. Calculate the resultant Force vector $\mathbf{F}$.

36) A circular loop of wire with a radius of 20 cm is placed in a uniform magnetic field of 0.5 Tesla. The plane of the loop makes an angle of 30 degrees with the magnetic field lines. Calculate the magnetic flux passing through the loop. (Be careful with which angle you use).

37) A rectangular coil with an area of 0.1 square meters is rotating at a constant angular velocity of ω radians/second inside a uniform magnetic field $B = 2.0$ Tesla. At time $t=0$, the coil is perpendicular to the field (maximum flux).

a) Write the equation for the magnetic flux $\Phi(t)$ as a function of time.

b) Describe the shape of the graph of Flux vs. Rotation Angle from 0 to 360 degrees, and state the flux values at 90 degrees and 180 degrees.

38) A hollow cube with side length $L = 0.5$ meters is placed in a uniform magnetic field $B = 1.5$ Tesla pointing in the positive x-direction. Two faces of the cube are perpendicular to the field, and the other four are parallel to it. Calculate the net magnetic flux leaving the entire closed surface of the cube.

39) A square loop of side length 2 meters lies in the xy-plane. A uniform magnetic field is not aligned with any axis, but is given by the vector $\mathbf{B} = (3 \mathbf{i} + 4 \mathbf{j} + 5 \mathbf{k})$ Tesla. Calculate the magnetic flux through the square loop. (Hint: Determine the area vector $\mathbf{A}$ first).

40) A square loop of wire (side length L) is entering a region of uniform magnetic field $\mathbf{B}$ at a constant velocity $\mathbf{v}$. The field is perpendicular to the loop.

- a) As the loop enters the field, does the flux increase linearly, quadratically, or sinusoidally with time?
- b) Once the loop is fully inside the uniform field and continuing to move, what is the magnetic flux?

41) You have a high-quality step-up transformer with a turns ratio of 1:100. You connect the primary coil to a standard 12-volt car battery (DC) and wait for the system to stabilize. What is the steady-state voltage measured across the secondary coil?

42) An ideal transformer is used to step up a voltage from 120V to 12,000V (a factor of 100) to power a neon sign. If the neon sign draws 10 Watts of power on the secondary side, how much power is drawn from the source on the primary side, and does the current increase or decrease on the secondary side compared to the primary?

43) An audio amplifier expects to see a "load" of 100 Ohms to function correctly. However, you only have a 4 Ohm speaker. You decide to use an ideal transformer to match the impedance. Should the transformer have more turns on the amplifier side (primary) or the speaker side (secondary), and what should the turns ratio (N_p/N_s) be?

44) In a specific ideal transformer, the primary coil has 200 turns and is connected to a 120V AC source. If you were to measure the voltage across just one single loop of wire on the secondary coil (assuming the secondary has 500 total turns), what voltage would you measure for that single loop?

45) A rectangular coil is rotating at a constant speed inside a uniform magnetic field to generate electricity. At the exact moment when the coil is perpendicular to the magnetic field lines, the magnetic flux passing through the coil is at its absolute maximum. At this specific instant, is the induced EMF (voltage) maximum, minimum, or zero?

46) You are turning the handle of a hand-cranked AC generator to power a lightbulb. If you suddenly spin the handle twice as fast (doubling the frequency of rotation), what happens to the maximum peak voltage output? Does it stay the same, double, or quadruple?

47) A strong bar magnet is dropped vertically down a long copper tube. Usually, the magnet falls slowly due to magnetic braking (eddy currents). If you were to slice a thin vertical slit down the entire length of the tube (interrupting the circumference), how would the acceleration of the falling magnet compare to free-fall gravity?

48) A square loop of wire is moving at a constant velocity to the right through a region of uniform magnetic field that is much larger than the loop itself. While the entire loop is completely inside this uniform field region, is there an induced current flowing in the loop? Explain why or why not.

49) A metal rod is sliding to the right on frictionless rails, completing a circuit with a resistor. The setup is bathed in a magnetic field. To keep the rod moving at a constant velocity v , you must apply a constant force F . If you want the rod to move at twice the velocity, how much force must you apply compared to the original force F ? (Assume the resistor is the only component dissipating energy).

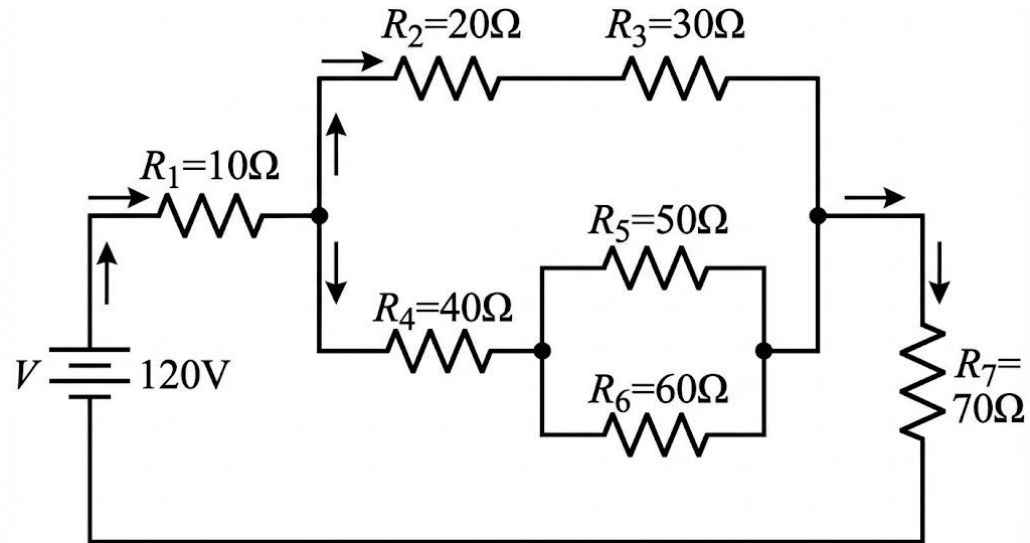
50) A long straight wire carries a current flowing upward. A rectangular loop of wire is placed to the right of the straight wire, lying flat in the same plane. If the current in the straight wire is suddenly decreased, in which direction (clockwise or counter-clockwise) will the induced current flow in the rectangular loop?

SOLUTIONS

- 1) 0.8 Ohms
- 2) 53,280 Joules; 20 Hours
- 3) 30,000 Amps; 3,000 Gigawatts
- 4) 72,000 Coulombs
- 5) 4,320 Megajoules
- 6) Yes, the fuse will blow.
- 7) Voltage = 20 V; Resistance = 10 Ohms
- 8) 1.2×10^{15} electrons; 0.00096 Joules
- 9) \$0.72
- 10) System B processes more charge; System B consumes more power.

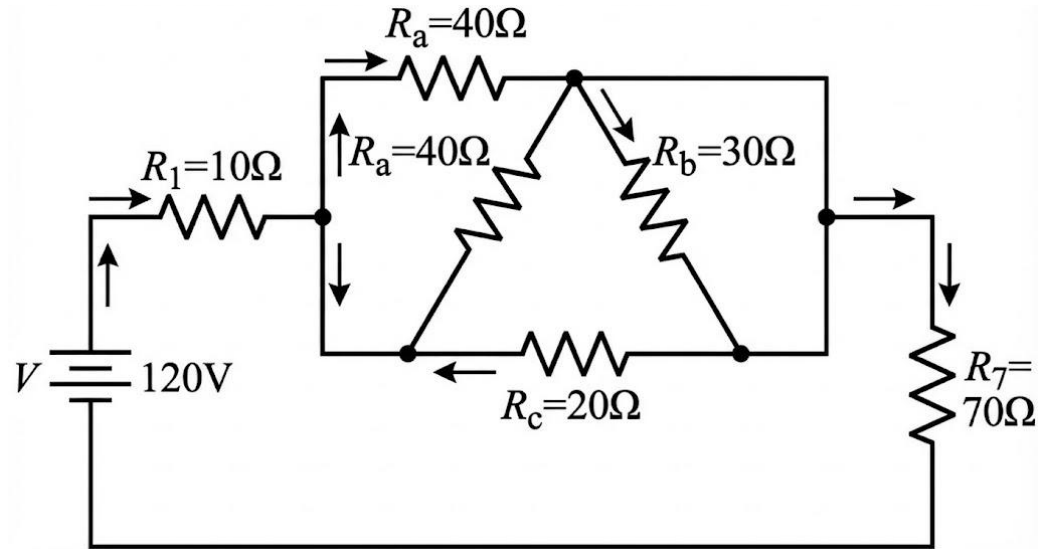
- 11) Find the equivalent resistance of this circuit
- 12) Find the power dissipated through resistor R_6

$$R_{eq} = 109 \text{ Ohms}$$
$$P = 2.81 \text{ W}$$



- 13) Find the equivalent resistance of this circuit
14) Find the power dissipated through resistor R_c

$$R_{eq} = 90 \text{ Ohms}$$
$$P = 8.89 \text{ W}$$

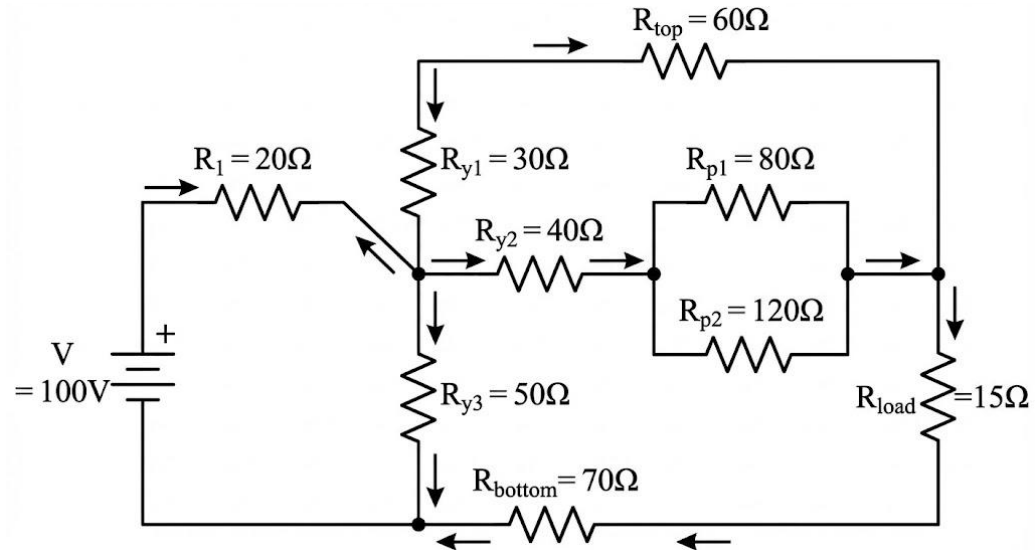


15) Find the equivalent resistance of this circuit

16) Find the current through resistor R_{y3}

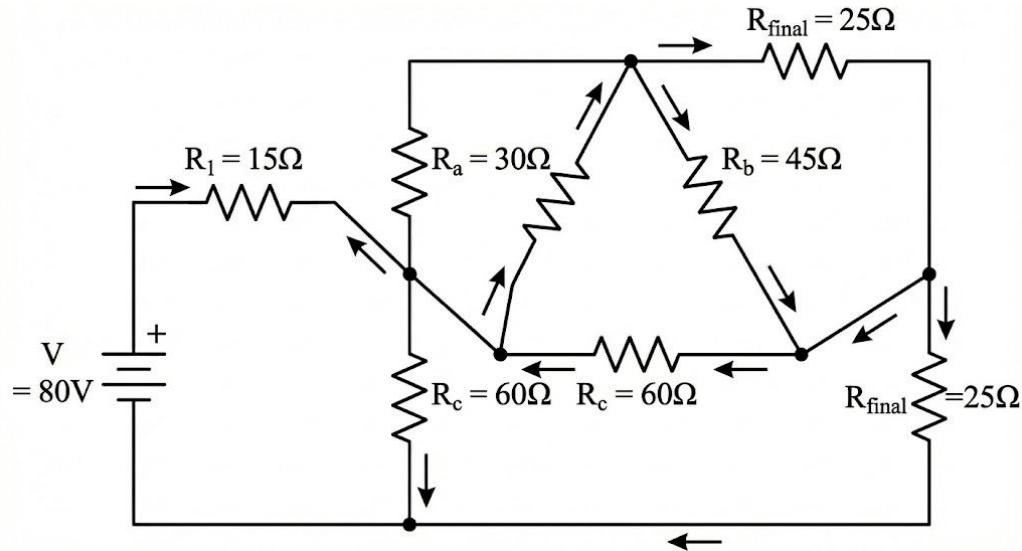
$$R_{eq} = 56 \text{ Ohms}$$

$$I = 1.284 \text{ A}$$



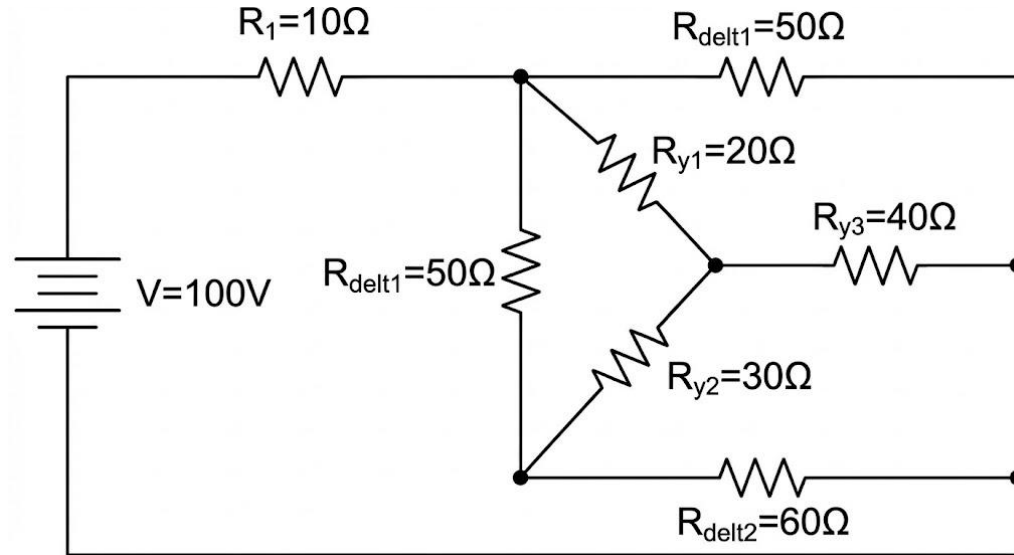
- 17) Find the equivalent resistance of this circuit
18) Find the power dissipated through resistor

$$R_{eq} = 40.867 \text{ Ohms}$$
$$P = 3.09 \text{ W}$$



- 19) Find the equivalent resistance of this circuit
20) Find the power dissipated through resistor R_{y2}

$$R_{eq} = 31.76 \text{ Ohms}$$
$$P = 0.4177 \text{ W}$$



- 21) $q = \text{square root of } [(4 * m * g * L^2 * \sin^2(\theta) * \tan(\theta)) / k]$
- 22) $k * x = (k_e * q^2) / (L_0 + x)^2$
- 23) h is approximately 1.05 meters
- 24) The charge q must be $-(9/16)Q$, placed at a distance of $d/4$ from the $+Q$ charge.
- 25) Kinetic Energy (K) = $(k * e^2) / (2 * r)$
- 26) East
- 27) It bends AWAY from the wire.
- 28) 0.25 meters from Wire A.
- 29) B
- 30) Top-Right (Quadrant 1) and Bottom-Left (Quadrant 3).

- 31) $F = 2.4 \times 10^{-17} \text{ j Newtons}$
- 32) 20 Newtons
- 33) East (positive i direction)
- 34) 3.92 Amps, flowing Left-to-Right
- 35) $F = -4.24 \text{ k Newtons}$
- 36) 0.0314 Webers
- 37) a) $\Phi(t) = 0.2 \cos(\omega t)$; b) Cosine wave, 0 Wb, -0.2 Wb
- 38) 0 Webers
- 39) 20 Webers
- 40) a) Linearly; b) Constant ($B \cdot L^2$)

- 41) 0 Volts
- 42) 10 Watts; The current decreases.
- 43) More turns on the Amplifier (Primary) side; Ratio is 5:1.
- 44) 0.6 Volts
- 45) Zero
- 46) It Doubles
- 47) Acceleration will be approximately equal to g (Free fall)
- 48) No current.
- 49) 2F
- 50) Clockwise

**Hopefully you don't want to kill me after these
exercises, it's for your sake :)
If you are able to do these, you'll do just fine in the
exam :)**